Maths By Gagan Pratap

Trigonometry Sheet-3

Maths Special Batch By Gagan Pratap

1. If $\cot\theta = \frac{80}{39}$, find the value of $cosec\theta$.

यदि $\cot\theta = \frac{80}{39}$,तो $cosec\theta$ का मान ज्ञात करे।

(a) $\frac{39}{80}$ (b) $\frac{89}{39}$

(c) $\frac{39}{89}$ (d) $\frac{89}{80}$

2. If $1 + \cot^2\theta = \frac{625}{49}$ and θ is acute, then what is the value of $(\sin\theta + \cos\theta)^{1/2}$?

यदि $1 + \cot^2\theta = \frac{625}{49}$ और θ न्यून कोण है, तो $(\sin\theta+\cos\theta)^{1/2}$ का मान क्या होगा?

(a) $\frac{\sqrt{3}}{5}$ (b) $\frac{\sqrt{31}}{7}$ (c) $\frac{\sqrt{31}}{5}$ (d) $\frac{\sqrt{33}}{8}$

3. If tan(x+y) = 1 and cos(x–y) = $\sqrt{3}/2$, then what is the value of x and y?

यदि tan(x+y) = 1 and cos(x–y) = $\sqrt{3}/2$, तो x और y का मान क्या है?

(a) x = 3, y = 4.5

(b) x = 37.5, y = 7.5

(c) X = 7.5, y = 37.5

(d) x = 4.5, y =3

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4. If secA = $\frac{5}{4}$, then the value of $\frac{tanA}{1+tan^2A} - \frac{sinA}{secA}$ is:

यदि secA = $\frac{5}{4}$, तो $\frac{tanA}{1+tan^2A} - \frac{sinA}{secA}$ का मान ज्ञात कीजिए।

(a) 2

(b) 1

(c) 0

(d) 3

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5. If x = asecⁿθ, y = b tanⁿθ is , then it will be ….

यदि **x = asecⁿθ, y = b tanⁿθ** है, तो ….. होगा।

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[A] $\left(\frac{x}{a}\right)^{2/n} + \left(\frac{y}{b}\right)^{2/n} = 1$ [B] $\left(\frac{a}{x}\right)^{2/n} + \left(\frac{b}{y}\right)^{2/n} = 1$

[C] $\left(\frac{x}{a}\right)^{2/n} - \left(\frac{y}{b}\right)^{2/n} = 1$ [D] $\left(\frac{a}{x}\right)^{2/n} - \left(\frac{b}{y}\right)^{2/n} = 1$

6. x If $x = a + aseca.cos\beta$, $y = b(1 + seca.sin\beta)$ and $z = c.tan\alpha$, then $\left(\frac{x-a}{a}\right)^2 + \left(\frac{y-b}{b}\right)^2 - \left(\frac{z}{c}\right)^2$=?

(a) 2 **(b) 1** **(c) – 1** **(d) 0**

7. If m = a sec A and y = b tan A, then find the value of $b^2m^2 - a^2y^2 + \frac{a^2y^2}{b^2m^2} + \cos^2A$.

यदि m = a sec A और y = b tan A है, तो $b^2m^2 - a^2y^2 + \frac{a^2y^2}{b^2m^2} + \cos^2A$ का मान ज्ञात कीजिए।

(a) a^2b^2 (b) $1-a^2b^2$

(c) a^2b^2+2 (d) a^2b^2+1

8. If $\sec\theta + cos\theta = 2$, find the value of $\sec^{55}\theta + \frac{1}{sec^{55}\theta}$.

यदि $\sec\theta + cos\theta = 2$ है, तो $\sec^{55}\theta + \frac{1}{sec^{55}\theta}$ का मान ज्ञात कीजिए।

(a) 2

(b) 0

(c) 1

(d) 55

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9. If $\tan\theta + \cot\theta = 2$, θ is an acute angle, then find the value of $2\tan^{25}\theta + 3\cot^{20}\theta + 5\tan^{30}\theta\cot^{15}\theta$.

यदि $\tan\theta + \cot\theta = 2$ है, θ एक न्यून कोण है, तो $2\tan^{25}\theta + 3\cot^{20}\theta + 5\tan^{30}\theta\cot^{15}\theta$ का मान ज्ञात कीजिए

(a) 12 (b) 10 (c) 8 (d) 6

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10. If $\tan\theta + \cot\theta = -2$, then the value of $\tan^9\theta + \cot^9\theta$ is:

यदि $\tan\theta + \cot\theta = -2$ है, तो $\tan^9\theta + \cot^9\theta$ का मान ज्ञात कीजिए।

(a) 2 (b) –1 (c) –2 (d) 0

11. Find the value of $\cot^2 B - \operatorname{cosec}^2 B$ for $0 < B < 90°$.

$0 < B < 90^o$ के लिए $cot^2B - cosec^2B$ का मान ज्ञात कीजिए।

(a) 2 (b) -1 (c) 0 (d) -2

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12. Simplify the given equation. $(1+\tan^2A)(1+\cot^2A) = ?$

दिए गए समीकरण को सरल कीजिए । $(1 + tan^2A)(1 + cot^2A) =?$

(a) $\frac{1}{\cos^2 A(1+\sin^2 A)}$ (b) $\frac{1}{\sin^2 A + \operatorname{cosec}^2 A}$

(c) $\frac{1}{\sin^2 A(1-\sin^2 A)}$ (d) $\frac{1}{\sin^2 A(1+\cos^2 A)}$

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13. Simplify:/सरल कीजिए:

$(1 + \tan^2A) + (1 + \frac{1}{tan^2A})$

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[A] $\frac{1}{sin^2A-sin^4A}$ [B] $\frac{1}{sin^2A-cos^4A}$

[C] $\frac{1}{cos^2A-sin^4A}$ [D] $\frac{1}{cot^2A-tan^4A}$

14. The value of $(\operatorname{cosec}^2 B - 1)(\sec^2 B - 1)$ is:

$(cosec^2B - 1)(sec^2B - 1)$ का मान क्या है?

(a) 3 (b) 4 (c) 1 (d) 2

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15. Simplify the given equation. $\frac{\cot^2 A - 1}{\cot A - 1} = ?$

दिए गए समीकरण को सरल कीजिए। $\frac{cot^3\ A-1}{cotA-1} = ?$

(a) $\operatorname{cosec}^2A + \cot A$ (b) $\operatorname{cosec}^2A - \cot A$ (c) $\cot^2A - \operatorname{cosec}A$ (d) $\cot^2A + \operatorname{cosec}A$

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16. Which of the following will satisfy $a^2 = b^2 + (ab)^2$ for the values a and b?

निम्नलिखित में से कौन a और b के लिए $a^2 = b^2 + (ab)^2$ संतुष्ट करेगा

A) a = sinx, b = cotx
B) a =cosx, b = tanx
C) a = cotx, b = cosx
D) a = sinx, b = tanx

17. $cot^2x - tan^2x$=?

a)4sec2xcosec2x b)4cot2xcosec2x c)sin8x d)4tan2xsec2x

18. If $2\tan x + 3\cot x = 7$, then the value of $4\tan^2x + 9\cot^2x$ is:

यदि $2\tan x + 3\cot x = 7$, है, तो $4\tan^2x + 9\cot^2x$ का मान ज्ञात कीजिए।

(a) 37
(b) 31
(c) 32
(d) 40

19. If $1 + \sin^2\theta = a\cos^2\theta$, then find $\sec^4\theta - \tan^4\theta$.

यदि $1 + \sin^2\theta = a\cos^2\theta$, तो $\sec^4\theta - \tan^4\theta$ ज्ञात कीजिए।

1. a^{-2}
2. a^{-1}
3. a
4. a^2

(SSC SELECTION POST XI 2023)

20. If $\sec^2A + \tan^2A = \frac{4}{17}$, $\sec^4A - \tan^4A$ is equal to:

यदि $\sec^2A + \tan^2A = \frac{4}{17}$ है, तो $\sec^4A - \tan^4A$ का मान क्या होगा?

(a) 13/17

(b) 4/13

(c) 4/17

(d) 5/17

(SSC CGL 2022)

21. If $\operatorname{cosec}^2\theta + \cot^2\theta = \frac{1}{3}$, where $0 \le \theta \le \frac{\pi}{2}$, then the value of $\cot^4\theta - \operatorname{cosec}^4\theta$ is:

यदि $\operatorname{cosec}^2\theta + \cot^2\theta = \frac{1}{3}$ है, जहाँ $0 \le \theta \le \frac{\pi}{2}$ है, तो $\cot^4\theta - \operatorname{cosec}^4\theta$ का मान........... है।

(a) 2/3

(b) –1/3

(c) 1/3

(d) –2/3

22. If $\tan\theta + \cot\theta = 4$, then the ratio of $3(\tan^2\theta + \cot^2\theta)$ to $(2\operatorname{cosec}^2\theta\sec^2\theta - 4)$ will be:

यदि $\tan\theta + \cot\theta = 4$ है, तो $3(\tan^2\theta + \cot^2\theta)$ का $(2\operatorname{cosec}^2\theta\sec^2\theta - 4)$ से अनुपात ज्ञात करें।

(a) 4 : 3

(b) 3 : 4

(c) 5 : 4

(d) 3 : 2

23. If $\sin t + \cos t = \frac{4}{5}$, then find $\sin t.\cos t$.

यदि $\sin t + \cos t = \frac{4}{5}$ है, तो $\sin t.\cos t$ का मान ज्ञात कीजिए।

(a) $\frac{9}{50}$ (b) $\frac{-9}{50}$ (c) $\frac{9}{50}$ (d) $\frac{-9}{50}$

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24. If $\sin A - \cos A = \frac{\sqrt{3}-\sqrt{2}}{2}$, then the value of $\sin A.\cos A$ is:

यदि $\sin A - \cos A = \frac{\sqrt{3}-\sqrt{2}}{2}$, तो $\sin A.\cos A$ का मान ज्ञात कीजिए।

(a) $\frac{2\sqrt{3}-1}{4}$ (b) $\frac{2\sqrt{2}-1}{8}$

(c) $\frac{2\sqrt{6}-1}{8}$ (d) $\frac{3\sqrt{2}-1}{4}$

25. If $\sin\theta + \cos\theta = 1.25$, then find the value of $\tan\theta + \cot\theta$?

यदि $\sin\theta + \cos\theta = 1.25$ है, तो $\tan\theta + \cot\theta$ का मान ज्ञात कीजिये?

A) $3\frac{5}{9}$

B) 2.666

C) 0.8

D) $\frac{25}{16}$

26. If $\sin\theta + \cos\theta = \frac{\sqrt{3}-1}{2\sqrt{2}}$, then what is the value of $\tan\theta + \cot\theta$?

यदि $\sin\theta + \cos\theta = \frac{\sqrt{3}-1}{2\sqrt{2}}$, तो $\tan\theta+\cot\theta$ का मान क्या है?

A) $8(\sqrt{3}-2)$

B) $12(\sqrt{3}-2)$

C) $12(\sqrt{3}+2)$

D) $8(\sqrt{3}+2)$

27. If $8\cos x - 8\sin x = 3$, then find $55\tan x + \frac{55}{\tan x}$?

यदि $8\cos x - 8\sin x = 3$ है, तो $55\tan x + \frac{55}{\tan x}$ ज्ञात कीजिये?

A) 128
B) 64
C) 55
D) None

28. If tan(t) + cot(t) = 16, then one of the values of the expression $\frac{1}{\sin(t)+\cos(t)}$ is _____ .

यदि tan(t) + cot(t) = 16, तो व्यंजक $\frac{1}{\sin(t)+\cos(t)}$ का एक मान ________ है।

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[A]0.75　　[B]$\frac{2}{\sqrt{3}}$

[C]6.25　　[D]$\frac{2\sqrt{2}}{3}$

29. Find the value of $3(\sec^2A-1)(\operatorname{cosec}A-1)(\operatorname{cosec}A+1)+4(\operatorname{cosec}^2A-\cot^2A)$?

$3(\sec^2A-1)(\operatorname{cosec}A-1)(\operatorname{cosec}A+1)+4(\operatorname{cosec}^2A-\cot^2A)$ का मान ज्ञात कीजिये?

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A) 10
B) 12
C) 3
D) 7

30. Consider the following:

i) $\sqrt{\sec^2\theta+\operatorname{cosec}^2\theta} = \tan\theta+\cot\theta, \text{where } 0° < \theta < 90°$

ii) $\sqrt{\tan^2\theta+\cot^2\theta} = \sec\theta+\operatorname{cosec}\theta, \text{where } 0° < \theta < 90°$

Which of the above are identities?

निम्नलिखित को धयान मे रखते हुए:

i) $\sqrt{\sec^2\theta+\operatorname{cosec}^2\theta} = \tan\theta+\cot\theta, \text{where } 0° < \theta < 90°$

ii) $\sqrt{\tan^2\theta+\cot^2\theta} = \sec\theta+\operatorname{cosec}\theta, \text{where } 0° < \theta < 90°$

A) 1 only
B) 2 only
C) Bothe 1 and 2
D) Neither 1 nor 2

31. If α is an acute angle, and $\tan^2\alpha - \sec\alpha = 19$, then find the value of cos α.

यदि α का एक न्यून कोण है, और $\tan^2\alpha - \sec\alpha = 19$ है, तो cos α का मान ज्ञात कीजिए। RRB Group D-2022

[A]$\frac{2}{3}$　　[B] $\frac{1}{5}$

[C]$\frac{1}{2}$　　[D] $\frac{1}{4}$

32. If A, B, C, D are the angles of a cyclic quadrilateral, then what is the value of $\sin\frac{A+C}{2}+\sin\frac{B+D}{2}$?

यदि A, B, C, D एक चक्रीय चतुर्भुज के कोण हैं, तो $\sin\frac{A+C}{2}+\sin\frac{B+D}{2}$ का मान क्या है?

A) 2　　C) 1
B) 0　　D) -1

33. If A, B, C are acute angles and sin (B+C-A) = cos(C+A-B)=tan(A+B-C)=1, then what is (A+B+C) equal to?

यदि A, B, C न्यून कोण हैं और sin (B+C-A) = cos(C+A-B)=tan(A+B-C)=1 है, तो (A+B+C) किसके बराबर है?

A) 90°　　c) 120°
B) 135°　　d) 150°

34. If $(\sec\alpha+\tan\alpha)(\sec\beta-\tan\beta)(\sec\gamma+\tan\gamma)(\sec\delta-\tan\delta) = (\sec\alpha-\tan\alpha)(\sec\beta+\tan\beta)(\sec\gamma-\tan\gamma)(\sec\delta+\tan\delta)$, then value of each side = ?

a) ± 1
b) $\sec\alpha\sec\beta\cdot\sec\gamma\sec\delta$

c) $\tan\alpha\cdot\tan\beta\cdot\tan\gamma\cdot\tan\delta$
d) 0

35. Find the value of $4(sec^6\theta - tan^6\theta) - 6(sec^4\theta + tan^4\theta) + 5 = ?$

(a) 2 (b) 3 (c) 4 (d) 1

36. $(cot\theta + tan\theta)(cosec\theta - sin\theta)(cos\theta - sec\theta) = ?$ SSC CHSL PRE 2024

A) 0
B) 2
C) 1
D) -1

37. If $\tan 9^0 = \frac{p}{q}$, then the value of $\frac{sec^2 81^0}{1+cot^2 81^0}$ is

a) $\frac{p}{q}$ b) $\frac{p^2}{p^2+q^2}$ c) $\frac{p^2}{q^2}$ d) $\frac{q^2}{p^2}$

38. Find the value of $(\tan^2\theta + \tan^4\theta)$.

$(\tan^2\theta + \tan^4\theta)$ का मान ज्ञात कीजिए । (SSC CPO 2023)

(a) $Sec^2\theta - Sec^4\theta$
(b) $Cot^2\theta - tan^4\theta$
(c) $Sec^4\theta + Sec^4\theta$
(d) $Sec^2\theta - Sec^2\theta$

39. $\frac{1}{cosecA-cotA} - \frac{1}{sinA} = ?$

(a) $\frac{1}{sinA} + \frac{1}{cosecA+cotA}$
(b) $\frac{1}{sinA} - \frac{1}{cosecA+cotA}$
(c) $\frac{1}{sinA} - \frac{1}{cosecA-cotA}$
(d) $\frac{1}{sinA} + \frac{1}{cosecA-cotA}$

40. If $tan^2 a = 3 + Q^2$, then $sec\ a + tan^3 a\ cosec\ a = ?$

$tan^2 a = 3 + Q^2$ है, तो $sec\ a + tan^3 a\ cosec\ a = ?$

(a) $(3+Q^2)^{3/2}$
(b) $(7+Q^2)^{3/2}$
(c) $(5-Q^2)^{3/2}$
(d) $(4+Q^2)^{3/2}$ (SSC CGL 2022)

41. $[\frac{1}{secA+tanA}]^2 = ?$

a) secA +tanA b) sinA cosA
c) $\frac{1-sinA}{1+sinA}$ d) $\frac{1-cosA}{1+cosA}$

42. Evaluate the following expression in terms of trigonometric ratios.

निम्नलिखित का मान त्रिकोणमितीय अनुपातों के पदों में ज्ञात करें।

$$\frac{secA - tanA}{secA + tanA}$$

(a) $1 + 2\tan^2 A + 2secA\ tanA$
(b) $1 + 2\sec^2 A - 2secA\ tanA$
(c) $1 + 2\tan^2 A - 2secA\ tanA$
(d) $1 + 1\sec^2 A + 2secA\ tanA$

43. $1 + 2\cot^2\theta + 2\cos\theta\ cosec^2\theta$, $0° < \theta < 90°$, is equal to :

$1 + 2\cot^2\theta + 2\cos\theta\ cosec^2\theta$, $0° < \theta < 90°$ का मान ज्ञात करें।

(a) $\frac{1-\sin\theta}{1+\sin\theta}$

(b) $\frac{1+\cos\theta}{1-\cos\theta}$

(c) $\dfrac{1-\cos\theta}{1+\cos\theta}$

(d) $\dfrac{1+\sin\theta}{1-\sin\theta}$

44. If $1+2tan^2\theta+2sin\theta sec^2\theta=\frac{a}{b}, 0^0<\theta<90^0$, then $\frac{a+b}{a-b}=?$

यदि $1+2tan^2\theta+2sin\theta sec^2\theta=\frac{a}{b}, 0^0<\theta<90^0$, तो $\frac{a+b}{a-b}=?$

(a) $sin\theta$ (b) $cos\theta$ (c) $cosec\theta$ (d) $sec\theta$

45. If $27\,sec^2\,\theta-11\,tan^2\,\theta=52$, Then $\frac{1+2\sin\theta\cdot\cos\theta}{1-2\sin\theta\cdot\cos\theta}=?$

(a)90 (b)81

(c)76 (d)63

46. $x=2y\cot\theta+z, z=x\cos\theta-3y$ then which one is true?

a. $1+\frac{4y^2}{(z-x)^2}=\left(\frac{z+3y}{x}\right)^2$

b. $1+\frac{4y^2}{(x-z)^2}=\frac{x^2}{(z+3y)^2}$

c. $1-\frac{x}{(z+3y)^2}=\frac{-4y^2}{(x-z)^2}$

d. $1+\left(\frac{2y}{x-z}\right)^2=\frac{x}{(z+3y)^2}$

47. What is the value of $\frac{1+tanA}{cosecA}+\frac{1+cotA}{secA}$?

$\frac{1+tanA}{cosecA}+\frac{1+cotA}{secA}$ का मान क्या है?

(a) $2\sec^2 A$

(b) sec A + cosec A

(c) sec A + cosec A

(d) $2\,cosec^2$

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48. $\left[1+\frac{tan^2A}{1+secA}\right]\left[\frac{cot^2A}{cosecA-1}-1\right]=?$

a)2cos2A b)2sin2A

c)2cosec2A d)2tan2A

49. $\frac{cotx}{1+cosecx}+\frac{1+cosecx}{cotx}$ is equal to:

$\frac{cotx}{1+cosecx}+\frac{1+cosecx}{cotx}$ का मान ____ के बराबर होगा।

(a) 2 sec x (b) 2 cos x

(c) 2 cosec x (d) 2 sin x

50. Simplify: $1+\frac{tan^2\emptyset}{sec\emptyset+1}+\frac{cosec\theta+1}{cot\theta}-\frac{cot\theta}{cosec\theta-1}$

(a) 1 (b) $sec\emptyset$

(c) $cosec\theta$ (d) $sin\theta cos\emptyset$

51. The expression $3cosec^2\theta\,cot^2\theta+cot^6\theta-cosec^6\theta$ is equal to : –

a) -2 b) 1 c) 2 d) -1

52. The value of $\frac{sec^6\theta-tan^6\theta-3sec^2\theta tan^2\theta+1}{cos^4\theta-sin^4\theta+2sin^2\theta+2}$ is?

$\frac{sec^6\theta-tan^6\theta-3sec^2\theta tan^2\theta+1}{cos^4\theta-sin^4\theta+2sin^2\theta+2}$ का मान ज्ञात कीजिये ?

a) $\frac{3}{4}$ b) $\frac{2}{3}$ c) $\frac{1}{2}$ d) 1

53. The value of $\left(\frac{sin\theta}{1+cos\theta}+\frac{1+cos\theta}{sin\theta}\right)\left(\frac{1}{tan\theta+cot\theta}\right)$ is equal to?

$\left(\frac{sin\theta}{1+cos\theta}+\frac{1+cos\theta}{sin\theta}\right)\left(\frac{1}{tan\theta+cot\theta}\right)$ का मान किसके बराबर है? SSC CHSL PRE 2024

A) $Tan\theta$

B) 1

C) $2cos\theta$

D) $2sin\theta$

54. $\frac{(\sec\theta+\tan\theta)(1-\sin\theta)}{\operatorname{cosec}\theta(1+\cos\theta)(\operatorname{cosec}\theta-\cot\theta)}$ is equal to:

$\frac{(\sec\theta+\tan\theta)(1-\sin\theta)}{\operatorname{cosec}\theta(1+\cos\theta)(\operatorname{cosec}\theta-\cot\theta)}$ बराबर है:

a) $\sin\theta$
b) $\sec\theta$
c) $\cos\theta$
d) $\operatorname{cosec}\theta$

55. $\frac{\sin\theta[(1-\tan\theta)\tan\theta+\sec^2\theta]}{(1-\sin\theta)\tan\theta(1+\tan\theta)(\sec\theta+\tan\theta)}$ is equal to:

$\frac{\sin\theta[(1-\tan\theta)\tan\theta+\sec^2\theta]}{(1-\sin\theta)\tan\theta(1+\tan\theta)(\sec\theta+\tan\theta)}$ ________ के बराबर है।

(a) 1
(b) cosecθ secθ
(c) sinθ cosθ
(d) – 1

56. The value of $\frac{\sec^2\theta(2+\tan^2\theta+\cot^2\theta)\div(\sin^2\theta-\tan^2\theta)}{(\operatorname{cosec}^2\theta+\sec^2\theta)(1+\cot^2\theta)^2}$ is:

$\frac{\sec^2\theta(2+\tan^2\theta+\cot^2\theta)\div(\sin^2\theta-\tan^2\theta)}{(\operatorname{cosec}^2\theta+\sec^2\theta)(1+\cot^2\theta)^2}$ का मान ज्ञात कीजिए।

(a) – 1
(b) 1
(c) – 2
(d) 2

57. The expression $\frac{\tan^6\theta-\sec^6\theta+3\sec^2\theta\tan^2\theta}{\tan^2\theta+\cot^2\theta+2}$, 0°<θ<90° , is equal to:

व्यंजक $\frac{\tan^6\theta-\sec^6\theta+3\sec^2\theta\tan^2\theta}{\tan^2\theta+\cot^2\theta+2}$ का मान बताइए जहां 0°<θ<90° है।

(a) $\sec^2\theta\operatorname{cosec}^2\theta$
(b) $-\sec^2\theta\operatorname{cosec}^2\theta$
(c) $\cos^2\theta\sin^2\theta$
(d) $-\cos^2\theta\sin^2\theta$

58. The expression $\frac{\cos^4\theta-\sin^4\theta+2\sin^2\theta+3}{(\operatorname{cosec}\theta+\cot\theta+1)(\operatorname{cosec}\theta-\cot\theta+1)-2}$, 0°<θ<90° , is equal to

व्यंजक $\frac{\cos^4\theta-\sin^4\theta+2\sin^2\theta+3}{(\operatorname{cosec}\theta+\cot\theta+1)(\operatorname{cosec}\theta-\cot\theta+1)-2}$, का मान बताइए, जहां 0°<θ<90° है।

(a) $\frac{1}{2}\sin\theta$ (b) 2sinθ (c) secθ (d) 2cosecθ

59. If $\frac{(\cos^6\theta+\sin^6\theta-1)(\tan^2\theta+\cot^2\theta+2)+2}{(\sec\theta+\tan\theta)(\sec\theta-\tan\theta)^2\cos\theta}=\frac{1}{k-1}$, $0°<\theta<90°$, then find k?

यदि $\frac{(\cos^6\theta+\sin^6\theta-1)(\tan^2\theta+\cot^2\theta+2)+2}{(\sec\theta+\tan\theta)(\sec\theta-\tan\theta)^2\cos\theta}=\frac{1}{k-1}$, $0°<\theta<90°$ है, तो k ज्ञात करें? (ICAR Assistant 2022)

A) $\text{Sin}\theta$
B) $\text{Cos}\theta$
C) $\text{Sec}\theta$
D) $\text{Cosec}\theta$

60. Who will be equal to $\frac{(\sin\theta-\cos\theta)(1+\tan\theta+\cot\theta)(\operatorname{cosec}^2\theta)}{\sec^3\theta-\operatorname{cosec}^3\theta}$, $0°<\theta<90°$?

$\frac{(\sin\theta-\cos\theta)(1+\tan\theta+\cot\theta)(\operatorname{cosec}^2\theta)}{\sec^3\theta-\operatorname{cosec}^3\theta}$, $0°<\theta<90°$ किसके बराबर होगा? (ICAR Technician 2022)

A) $\sin^2\theta$
B) $\sin\theta\cos\theta$
C) $\cos^2\theta$
D) $\sec\theta\operatorname{cosec}\theta$

61. Find the value of $\frac{\cos\theta(1+\sin\theta)}{(1+\sin\theta-\cos^2\theta)}\times\frac{(\tan\theta+\cot\theta)^{-1}}{(\operatorname{cosec}\theta-\sin\theta)(\sec\theta-\cos\theta)}$? (SSC CGL 2022)

$\frac{\cos\theta(1+\sin\theta)}{(1+\sin\theta-\cos^2\theta)}\times\frac{(\tan\theta+\cot\theta)^{-1}}{(\operatorname{cosec}\theta-\sin\theta)(\sec\theta-\cos\theta)}$ का मान ज्ञात कीजिये?

A) $\cot\theta$
B) $\sin^2\theta$
C) $\operatorname{cosec}^2\theta$
D) $\tan\theta$

62. Let $0^0<\theta<90^0$, $(1+\cot^2\theta)(1+\tan^2\theta)\times(\sin\theta-\operatorname{cosec}\theta)(\cos\theta-\sec\theta)$ is equal to:

मान लें कि $0^0<\theta<90^0$ है तो $(1+\cot^2\theta)(1+\tan^2\theta)\times(\sin\theta-\operatorname{cosec}\theta)(\cos\theta-\sec\theta)$ का मान इनमें से किसके बराबर होगा?

(a) $\sec\theta\operatorname{cosec}\theta$
(b) $\sec\theta+\operatorname{cosec}\theta$
(c) $\sin\theta+\cos\theta$
(d) $\sin\theta\cos\theta$

63. $\frac{(1+sec\theta cosec\theta)^2(sec\theta-tan\theta)^2(1+sin\theta)}{(sin\theta+sec\theta)^2+(cos\theta+cosec\theta)^2}$, $0^0 < \theta < 90^0$, is equal to:

$\frac{(1+sec\theta cosec\theta)^2(sec\theta-tan\theta)^2(1+sin\theta)}{(sin\theta+sec\theta)^2+(cos\theta+cosec\theta)^2} = ?$, $0^0 < \theta < 90^0$

(a) $1-cos\theta$ (b) $1-\ sin\theta$ (c) $cos\theta$ (d) $sin\theta$

64. $\left(\frac{tan^3\theta}{sec^2\theta}+\frac{cot^3\theta}{cosec^2\theta}+2sin\theta cos\theta\right) \div (1+cosec^2\theta+tan^2\theta)$, $0^0 < \theta < 90^0$ is equal to:

$\left(\frac{tan^3\theta}{sec^2\theta}+\frac{cot^3\theta}{cosec^2\theta}+2sin\theta cos\theta\right) \div (1+cosec^2\theta+tan^2\theta)$, $0^0 < \theta < 90^0$ का मान इनमें से किसके बराबर होगा?

a) $sin\theta cos\theta$ (b) $cosec\theta$ (c) $sec\theta$ (d) $cosec\theta sec\theta$

65. $(tan^2\theta+cot^2\theta+2) \div \left(\frac{1}{1+cos\theta}+\frac{1}{1-cos\theta}\right)$, $0° < \theta < 90°$, is equal to?

$(tan^2\theta+cot^2\theta+2) \div \left(\frac{1}{1+cos\theta}+\frac{1}{1-cos\theta}\right)$, $0° < \theta < 90°$ किसके बराबर है?

A) $\frac{cosec^2\theta}{2}$ C) $2cosec^2\theta$

B) $\frac{sec^2\theta}{2}$ D) $2sec^2\theta$

66. If $\frac{(tan\theta-sec\theta+1)}{(tan\theta+sec\theta-1)}$secθ $=\frac{1}{k}$, then k = ?

(a) 1 + sinθ (b) 1– cosθ (c) 1+ cosθ (d) 1– sinθ

67. What is $\frac{cotA+cosec\ A-1}{cotA-cosec\ A+1}$ equal to?

(a) $\frac{1+sinA}{cosA}$ (b) $sinA$ (c) tan$\frac{A}{2}$ (d) cot$\frac{A}{2}$

68. If $\frac{tan\theta}{1-cot\theta}+\frac{cot\theta}{1-tan\theta}$ = 1 + k, then k=?

(a) cotθ + secθ (b) tanθ + cosecθ (c) tanθ + secθ (d) cosecθ secθ

69. If $tan\,\theta + cot\,\theta = x$, then $tan^4\,\theta + cot^4\,\theta = ?$

(a)$(x^3-3)^2+2$ (b) $(x^4-2x)+4$

(c)$x(x-4)+2$ (d) $x^2(x^2-4)+2$

70. if p = $sec\theta - tan\theta$ and q = $cosec\theta + cot\theta$ then what is p + q(p-1) equal to ?

यदि p = $sec\theta - tan\theta$ और q = $cosec\theta + cot\theta$ है तो p + q(p-1) क्या होगा ?

(A) – 1 (B) 0 (C) 1 (D) 2

71. Find the value of (1+cotA – cosecA)(1+tanA +secA) –3(sin^2A+cos^2A)?

(1+cotA – cosecA)(1+tanA +secA) –3(sin^2A+cos^2A) का मान ज्ञात कीजिये?

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A) -1
B) 1
C) 2
D) -2

72. The value of expression $(1+sec32°+cot58°)(1-cosec32°+tan58°)$ is?

व्यंजक का मान (1+sec32°+cot58°)(1–cosec32°+tan58°) है?

A) 1 C) 4

B) 2 D) 5

73. $\frac{(1+tan\theta+sec\theta)(1+cot\theta-cosec\theta)}{(sec\theta+tan\theta)(1-sin\theta)}$ is equal to:

$\frac{(1+tan\theta+sec\theta)(1+cot\theta-cosec\theta)}{(sec\theta+tan\theta)(1-sin\theta)}$ __________ के बराबर है।

(a) 2 secθ (b) secθ

(c) 2 cosecθ (d) cosecθ

74. Simplify the following expression $\frac{(sec\theta-cos\theta)(cot\theta+tan\theta)}{(1+tan\theta+sec\theta)(sec\theta)(1+cot\theta-cosec\theta)}$, $0° < \theta < 90°$?

निम्नलिखित व्यंजक $\frac{(sec\theta-cos\theta)(cot\theta+tan\theta)}{(1+tan\theta+sec\theta)(sec\theta)(1+cot\theta-cosec\theta)}$, $0° < \theta < 90°$ को सरल कीजिए?

(ICAR Assistant 2022)

A) $\frac{tan\theta}{2}$
B) $\frac{cot\theta}{2}$
C) $cosec\theta sec\theta$
D) $sin\theta cos\theta$

75. If $\tan^4x - \tan^2x = 1$, then the value of $\sin^4x + \sin^2x$ is:

यदि $\tan^4x - \tan^2x = 1$ है, तो $\sin^4x + \sin^2x$ का मान ज्ञात कीजिए:

(a) $\frac{3}{4}$ (b) $\frac{1}{2}$

(c) 1 (d) $\frac{3}{2}$

76. If $\cot^2\theta + \cot^4\theta = 7$, then the value of $7\sin^4\theta + \sin^2\theta$ is:

यदि $\cot^2\theta + \cot^4\theta = 7$ है, तो $7\sin^4\theta + \sin^2\theta$ का मान ज्ञात करें।

(a) 3

(b) 5

(c) 1

(d) 2

77. If $\sec^5x - \sec^3x = 1$ then $\cos^2x + \cos^5x = ?$

a) 1 b) 0 c) 3/2 d) 1/2

78. If $\sec^2x - \sec x = 1$, then $\tan^6x + 3\tan^{10}x - 3\tan^8x - \tan^{12}x + 4 = ?$

a) 4 b) 3 c) 5 d) 6

79. If $\text{cosec}\,\theta + \text{cosec}^2\,\theta = 1$ then $\cot^{12}\theta - 3\cot^{10}\theta + 3\cot^8\theta - \cot^6\theta = ?$

a) -2 b) -1 c) 0 d) 1

80. If $\cot^4 A + \cot^2 A = 1$, then find the value of $\cos^4 A - 3\cos^2 A$.

यदि $\cot^4 A + \cot^2 A = 1$ है, तो $\cos^4 A - 3\cos^2 A$ का मान ज्ञात कीजिए। (CHSL MAINS 2023)

[a] –1 [b] –2 [c] 2 [d] 1

81. If $0^0 < \theta < 90^0$ and $\text{cosec}\,\theta = \cot^2\theta$, then the value of expression $\text{cosec}^4\theta - 2\text{cosec}^3\theta + \cot^2\theta$ is equal to:

a) 2 b) 0 c) 1 d) 3

82. If $\sec\theta = 4x$ and $\tan\theta = \frac{4}{x}$, $(x \neq 0)$ then the value of $12(x^2 - \frac{1}{x^2})$ is:

यदि $\sec\theta = 4x$ और $\tan\theta = \frac{4}{x}$, $(x \neq 0)$ है तो $12(x^2 - \frac{1}{x^2})$ का मान है:

a) $\frac{1}{2}$

b) $\frac{2}{3}$

c) 1

d) $\frac{3}{4}$

83. If secA + tanA = 5, then secA – tanA = ?

यदि secA + tanA = 5, then secA – tanA = ?

(a) 5/2 (b) 1/5 (c) 5 (d) 2/5

84. If $\frac{1-\cos\theta}{\sin\theta} = \frac{2}{7}$, then what will be the value of $\frac{1+\cos\theta}{\sin\theta}$?

यदि $\frac{1-\cos\theta}{\sin\theta} = \frac{2}{7}$, तो $\frac{1+\cos\theta}{\sin\theta}$ का मान क्या होगा?

A) 5

B) 2/7

C) 3.5

D) 1/5

85. If sec x + tan x = 5 and cosec y – cot y = 1/3, then find the value of (sec x + cosec y) – (tanx–coty).

यदि sec x + tan x = 5 and cosec y – cot y = 1/3, तो (sec x + cosec y) – (tanx–coty) का मान ज्ञात कीजिए।

(a) 2.2

(b) 3.2

(c) 4.2

(d) 3.1 SSC CHSL TIER – I 2022

86. If secθ-tanθ = $\frac{x}{y}$, (0<x<y) and (0°<θ<90°), then sin θ is equal to:-

यदि secθ - tanθ = $\frac{x}{y}$, (0 < x < y) और 0° < θ < 90° है, तो sin θ का मान ज्ञात कीजिए:

a) $\frac{x^2+y^2}{2xy}$ b) $\frac{2xy}{x^2+y^2}$ c) $\frac{y^2-x^2}{x^2+y^2}$ d) $\frac{x^2+y^2}{y^2-x^2}$

87. If 1 + sinθ = mcosθ, then what is the value of sinθ?

यदि 1 + sinθ = mcosθ है तो sinθ का मान क्या होगा?

(a) $\frac{2m^2-1}{m^2+1}$

(b) $\frac{m^2-1}{m^2+1}$

(c) $\frac{m^2+1}{2m^2-1}$

(d) $\frac{2m^2-1}{m^2-1}$ (SSC CGL 2022)

88. If $tan\theta + sec\theta = 3, then\ find\ 3tan\theta + 9sec\theta$?

यदि tanθ+secθ=3, तो 3tanθ+9secθ ज्ञात कीजिये?

A) 15 C) 17
B) 19 d) 21

89. If $sec\,\theta + tan\,\theta = \sqrt{21 + \sqrt{21 - \sqrt{21 + \sqrt{21 - \ldots\infty}}}}$

Then $sin\,\theta + cos\,\theta$ =?

(a)17/13 (b) 17/21
(c)13/21 (d) 21/13

90. If cosecθ – cotθ = m , then what is cosecθ equal to ?

यदि cosecθ – cotθ = m है तो cosecθ किसके बराबर होगा ?

(A) $m + \frac{1}{m}$ (B) $m - \frac{1}{m}$ (C) $\frac{m}{2} + \frac{2}{m}$ (D) $\frac{m}{2} + \frac{1}{2m}$

91. If cosecθ - cotθ = $\frac{7}{2}$, then value of sinθ=?

a)$\frac{47}{28}$ b)$\frac{28}{51}$ c)$\frac{28}{53}$ d)$\frac{28}{49}$

92. If secθ = x +$\frac{1}{4x}$ then $\sqrt{\frac{1+sin\theta}{1-sin\theta}}$ = ?

(a)2x , -$\frac{1}{2x}$ (b) 2x , $\frac{1}{2x}$
(c)2x (d) $\frac{1}{2x}$

93. If $\frac{1+sin\emptyset}{1-sin\emptyset} = \frac{p^2}{q^2}$, then $sec\emptyset$ is equal to:

यदि $\frac{1+sin\emptyset}{1-sin\emptyset} = \frac{p^2}{q^2}$, तो $sec\emptyset$ बराबर है :

a) $\frac{p^2q^2}{p^2+q^2}$ b) $\frac{1}{2}\left(\frac{q}{p} + \frac{p}{q}\right)$
c) $\frac{2p^2q^2}{p^2+q^2}$ d) $\frac{1}{p^2} + \frac{1}{q^2}$

94. If $sec\,\theta + tan\,\theta = P$, then $\frac{P^2+1}{P^2-1}$ =?

(a)$sin\,\theta$ (b) cosecθ
(c)$tan\,\theta$ (d) $cot\,\theta$

95. If a SecA + b tanA = c, then a tanA +b secA is equal to:

यदि a SecA + b tanA = c है, तो a tanA + b secA का मान क्या है?

(a) $\pm\sqrt{c^2 - b^2 + a^2}$ (b)$\pm\sqrt{a^2 + b^2 - c^2}$
(c) $\pm\sqrt{a^2 + b^2 + c^2}$ (d) $\pm\sqrt{c^2 - a^2 + b^2}$

96. If 29 secθ–21 tanθ = 20 then 29 tanθ – 21 secθ = ?

(a)0 (b)29
(c)21 (d)20

97. 5secθ – 4tanθ = 3cosecθ, then find the value of $\frac{27cot\theta}{5tan\theta-4sec\theta}$ =?

a) 5 b) 9 c) 7 d) 6

98. If $(x^2+y^2)\sec\theta + (x^2-y^2)\tan\theta = 9$ and $(x^2+y^2)\tan\theta + (x^2-y^2)\sec\theta = 5$, then $3x^2y^2 = ?$

a) 36 b) 42 c) 39.75 d) 48

99. If $5 + 12tan\theta = sec\theta$ & $12 - 5tan\theta = ksec\theta$, then find k^2?

यदि $5 + 12tan\theta = sec\theta$ & $12 - 5tan\theta = ksec\theta$ है, तो k^2 ज्ञात कीजिये?

A) 101
B) 27
C) 168
D) 99

100. If $a^2 cosec^2\theta - b^2 cot^2\theta = 7$ and $a\ cosec\ \theta + b\cot\theta = 1$, then $a^2(b^2+9) = ?$

(a) $14\ b^2$ (b) $16\ b^2$
(c) $25\ b^2$ (d) $10\ b^2$

101. If $\sec\theta$ and $\sin\theta (0 < \theta < 90)$ are the roots of the equation $\sqrt{6}x^2 - kx + \sqrt{6} = 0$, then the value of k is:

यदि secθ और sinθ, (0 <90 <90), समीकरण $\sqrt{6}x^2 - kx + \sqrt{6} = 0$ की जड़ें हैं, तो "k" का मान है:

(a) $\sqrt{3}$ (b) $3\sqrt{2}$ (c) $2\sqrt{3}$ (d) $3\sqrt{3}$